

Magitech

TOWERS OF SAEROTH · PLAYER HANDOUT

Machinery built to hold a working, and workings designed around what a machine can do that a hand cannot. Both halves are old; putting them together deliberately is not, and the people doing it have not agreed on what to call it. **Magitech** is the word that has stuck outside the trade, mostly because it is short.

Two ends of the same problem

Melisor Magocracy came at it from the magic. An academy that can already build a teleportation circle wants to know what a circle can be made to do when it is a component rather than a destination, and Melisor has the theorists, the ley-lines and the funding to ask.

Silicar came at it from the machine. A guild that has spent four generations on locks, gearing and irrigation wants to know which of its mechanisms get better with a working inside them, and Silicar has the workshops, the clockwork trade and a habit of building the thing before arguing about it. It is also the guild that invented the firelock, by accident, and has never once let that trade and this one share a bench. See Technology.

Neither of them finds this funny. Everybody else does.

Materials and power

A magitech device is built, not grown out of a spellbook, and it looks it. The good ones use metals and woods that got rare the moment the trade started buying them up, finished the way a luthier finishes a violin rather than the way a smith finishes a plough, and set with a gemstone or crystal more often than not. The metal itself is usually an alloy — a single pure metal holds a working's shape badly, and every workshop guards its own blend like a vintner guards a cellar.

Power is the harder problem, and neither academy solved it from nothing. An aeon stone in this trade is one gem carrying two separate cuts, not a battery wired to a separate working. The **charge rune** comes first and pulls raw magical energy through the gem, and the gem is not an interchangeable vessel for that: some stones catalyse the draw cleanly and some waste half of what the rune is trying to pull through them, which is why a workshop pays for a good ruby and not merely a red one. The **property rune** comes second, into the same stone, and shapes whatever the charge rune drew — and it doesn't take to every gem the same way either. Fire cuts cleanly into ruby and badly into almost anything else; a ward or open air takes to diamond; earth wants peridot. Cut the right rune into the wrong stone and the device is weak or doesn't hold at all, which is most of why the trade's pairings look fixed rather than arbitrary. Twenty years ago nobody in either academy could cut either rune, let alone say which stone suited which. What they had were whole low-level stones pulled out of old ruins, enough of them, studied long enough, to work both cuts back out from the finished article — and to notice, eventually, that the ruins had never once wasted a ruby on anything but fire.

The same ruins gave up broken housings too: alloy and wood still carrying a third cut, **circuitry**, carved into the material itself rather than the stone, that once carried a stone's shaped power the rest of the way into whatever the housing was built to do. That is where the trade got its wiring without inventing it a third time.

Property runes keep turning up in new varieties, and the trade keeps copying or inventing past them — fire, earth, air, a ward, a solvent, a season, with no sign the list is finished. The charge rune never has. Every stone anyone has ever pulled from a ruin carries the same basic cut, and in twenty years nobody has managed to design, or find, a more advanced one. A better stone or a cleaner alloy can close the gap between a device and its theoretical best, but neither can raise the ceiling itself. That ceiling is one specific cut that nobody has learned to improve. Whether that is a limit

of what survived the ruins or a limit neither academy has cracked on its own, Melisor and Silicar give two different answers in public and privately admit to the same one: they don't know.

Not everything that ships is a find. A few pieces are close reconstructions of something pulled half-whole out of a ruin and finished in a workshop; the rest are new work, designed from scratch around the recovered runework and the recovered circuitry, that happened to solve a problem the empire had already solved once. Nobody building either kind knows which one they're holding, because nobody building either kind knows there is a difference to ask about.

The list below names each device by its property rune only — the charge rune riding under it goes unmentioned the way nobody points out the nails in a chair. Twenty years in, Silicar's workshops have shipped a handful of things:

- **The wayfinder** — a disc small enough to wear on a cord, pointing toward whatever its stone was keyed to find. The one every court now wants and almost none can buy; Silicar's guild fills perhaps thirty orders a year.
- **The cold lamp** — no flame, no oil, no working laid on fresh each evening. A stone behind hammered glass, burning level and cold until it runs dry.
- **The strike-stone** — a ruby cut with a fire rune, set into a case of alloy the size of a thumb. Click it and it lights. No oil, no flint, no wick, which makes it the one magitech item an ordinary household might actually own, on the rare year it can find one to buy.
- **The slate** — a peridot cut with an earth rune, bedded into a tablet of worked stone. Write on it, press the rune, and the mark is gone — a scribe's whole complaint against parchment, solved. Thurion Merchant Alliance's counting-houses are already trying to buy the patent out from under Silicar before a rival house does.
- **The diving mask** — a diamond cut with an air rune, set into the glass. A salvage crew wearing one can work a wreck for the better part of an hour on a single charge, which has already changed what a reef-city considers worth the dive.

Every one of them is rare, every one is expensive, and every one works exactly as many times as its stones allow, and not once more.

Charge, pattern, impedance

Every apprentice guesses the same thing in their first week at a bench: that the stone, the cut and the metal all multiply together, and more of any one makes a better device. Three things do matter. It isn't the word *times* — though the shorthand chalked over every bench in Brightfurrow looks close enough to it:

$$W = Q \cdot P \div Z$$

Charge (Q) comes from the charge rune, but the gem it's cut into still matters. The rune sets a firm ceiling — no cut anyone has recovered pulls more than a low-level draw — and the gem decides how close to that ceiling a given stone actually gets. Some stones catalyse cleanly and some waste half of what the rune is trying to pull through them, which is why a clean ruby costs more than a merely red one even cut with the identical rune. Every charge rune anybody has ever recovered from a ruin is the same basic cut, and in twenty years nobody has managed to design a more advanced one. The ceiling on the whole trade is this one rune. What a good stone buys is getting near it.

Pattern (P) is the property rune, cut second into the same stone, and it has a ceiling of its own: how well it takes to the gem underneath it. Fire cuts cleanly into ruby and badly into almost anything else; a ward or open air takes to diamond; earth wants peridot. Cut a rune into a stone with no affinity for it and the device is weak or doesn't hold at all, which is most of why the trade's pairings look fixed rather than arbitrary. Because the cut is made once, when the stone itself is made, there is no reworking it after. Nobody reprograms an aeon stone. They requisition a new one, in the right gem this time, and cut it again.

Impedance (Z) is the circuitry — a third cut, carved not into the stone but into the alloy or the wood that houses it, carrying the shaped power the rest of the way out and refining it again on the way. Charge and Pattern both have a ceiling a good stone can approach but never clear. Impedance has none. Nobody has ever worked out a limit to how good a circuit can get, which is the real reason every guild in Brightfurrow guards its own circuitry patterns and alloy blends rather than publishing either. Bad impedance doesn't just waste charge as heat and backlash, burning through the stone in a third the uses a well-carved circuit would — it refines the effect badly too, which is the difference between a warded clasp that holds a clean half-shield and one that flickers.

The trade treats that chalked-up version as a rough thing that earns its keep at a bench where nobody has time for better. It takes one activation as a single lump of charge spent all at once, which is not what happens inside the housing. What happens is a stone bleeding charge over the fraction of a second the working takes, and a rate is not a lump:

$$w(t) = -(dQ/dt) \cdot P(t) \div Z(t)$$

The device's output at any instant is however fast the stone is losing charge right then, shaped by the pattern and cut down by whatever is in the way. None of that changes what a stone can afford across its whole working life, because a rate integrates back into a total the same way it always has:

$$\int_0^T w(t) dt \leq Q(0)$$

Spend it fast in one dramatic burst, or slow across a hundred small pulls, and the total is bounded the same either way. That is the actual shape of "works exactly as many times as its stone allows, and not once more" — not a count of uses at all, but an area under a curve that happens, for a well-made device with a steady draw, to look enough like a count that nobody in the trade has ever bothered to say it differently.

Put the three variables together, chalked or worked in full, and the trade's whole twenty years comes down to one number nobody can move, one number nobody can invent past, and one number that is the entire difference between a wayfinder and a toy. Melisor Magocracy has never had a reason to chase the third one. A warded clasp cut at Thelemar runs dry in a fraction of the uses a Silicar lamp gets from the same grade of stone, and nobody on the Council has ever once complained about it.

Neither academy has ever set any of this out as a body of law, which is not the same as nobody having done it. See Thaumodynamics.

Prestige pieces

Melisor Magocracy never needed magitech to solve anything. An academy that can put three teleportation circles under one campus has never once lacked a practical answer, so its half of the trade was never aimed at a problem. It was aimed at whoever already has every option and wants to be envied for the one they picked.

- **The warded clasp** — a diamond cut with an abjuration rune, pinned at the collar. It raises the same half-shield a first-year apprentice learns to cast, and holds it just as well, which has made it wildly popular with nobles who want to duel without earning the privilege the ordinary way. Ashwin Devaraj, who has spent a career being paid for that privilege and cited for none of it, has opinions about the clasp and states them loudly, usually in a tavern.
- **The presser** — a topaz cut with a cleansing rune, set into a flat iron. A launderer does the same work for a fraction of the price and cannot be shown off at a dinner party, which is rather the point of owning one instead.
- **The seasoning dish** — a citrine cut with a transmutation rune, laid under the centrepiece at a Council banquet. No two academies season a dish the same way, and a Thelemar dinner argues its politics through the courses as much as through anyone's conversation.

None of it is cheaper than paying somebody to do the job. That was never the offer.

The stakes

- **It is the first field in a century where Melisor Magocracy is not obviously ahead.** Silicar is small, is not a great power, and is producing results the academies did not predict.
- **The Council of Archmages cannot rule on it,** because two academies claim the same process and the Council's horror of ruling on anything is a policy older than the dispute.
- **Everything it produces is unlicensed,** in the same way the firelock is unlicensed. See Technology.
- **It is expensive and it is visible.** A magitech workshop is a large building with a great deal of money in it and no guild old enough to have arranged its own protection.

You cannot repeat any of this. Not to the party, not on paper, not in stone.

Thaumodynamics

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Twenty years of Magitech and nobody has written down what the trade is actually doing. Melisor Magocracy has theorists who will not go near a bench and Silicar has benchwrights who will not go near a theory, and between the two of them the field has accumulated a great deal of practice and no account of itself. What follows is an attempt at one. It is not published, it would embarrass both academies if it were, and the name for the study is the author's own coinage.

The fundamental bench equation

$W = Q \cdot P \div Z$

Chalked over every bench in Brightfurrow, usually by somebody who could not derive it. It estimates the useful Working a single-stone apparatus produces across one full cycle:

Term	Name	What it is
W	Working	The useful magical effect the apparatus actually delivers
Q	Charge	Thaumic potential the stone's charge rune makes available
P	Pattern	How well the property rune shapes that charge into the effect wanted
Z	Impedance	The opposition the material circuit puts up as patterned charge passes through it

So $W = f(Q, P, Z)$, and each of the three has a law governing it.

Charge is set by the charge rune, which fixes the ceiling, and by the gem, which decides how near the ceiling a given stone comes: $0 \leq Q \leq Q_{max}$. Every charge rune anybody has recovered gives the same Q_{max} , which is the whole of why the trade is stuck where it is.

Pattern operates on whatever charge is available, giving the product $Q \cdot P$. It is an effectiveness, not an identity — *which* property a rune expresses comes from the rune, and P says only how much of that property survives the cutting.

Impedance depends on the housing rather than the stone: $Z = f(M, C, \tau)$, for material, circuit geometry, and the kind of Working being carried.

Zeroth Law: Thaumic Affinity

A pattern expresses itself only in proportion to its affinity with the stone it is cut into.

$0 \leq P \leq P_{max}$

It is numbered zeroth because the other three laws all use P and none of them can say what P is. A property rune is not equally effective in every gem; its geometry has to agree with the crystal it is cut into. Fire agrees with ruby. A ward, or open air, agrees with diamond. Earth wants peridot. Fire cut into a gem with no affinity for it gives a weak effect or an unstable one, and a perfect pairing approaches P_{max} without ever passing it.

This is why particular stones became associated with particular workings, and why that association is not a convention anybody chose.

First Law: Conservation of Charge

No apparatus performs more total Working than the charge its stone made available. How fast charge is spent can be changed. The total cannot.

$$\int_0^T \mathbf{w}(t) dt \leq Q(0)$$

where $\mathbf{w}(t)$ is the instantaneous rate of useful Working, T the moment the working ends, and $Q(0)$ the charge in the stone when it began.

A firelock discharges enormously over a fraction of a second. A lamp gives almost nothing for a very long time. Both are bounded by the same integral, and this is the formal statement of what the trade already says in plainer words: a device works as many times as its stone allows and not once more.

Second Law: Thaumic Loss

Nothing is transmitted without loss. Imperfect affinity and circuit impedance both cut down the share of spent charge that arrives as useful Working.

Charge leaves the stone at a rate $-(dQ/dt)$. Some arrives as Working and some is wasted, and the two account for all of it:

$$-(dQ/dt) = \mathbf{w}(t) + \ell(t)$$

The share that arrives is the **conversion efficiency**, which is what the ratio of Pattern to Impedance has been measuring all along:

$$\eta(t) = P(t) \div Z(t), \text{ with } 0 \leq \eta \leq 1$$

$$\mathbf{w}(t) = \eta(t) \cdot (-dQ/dt) \text{ and } \ell(t) = (1 - \eta(t)) \cdot (-dQ/dt)$$

Waste leaves as heat, light, vibration, backlash, degradation of the stone itself, or an effect nobody asked for. A badly made lighter that draws ten units of charge to deliver six has not lost the other four. It has put them somewhere, and a workshop that cannot say where has a fire hazard rather than a device. This is the whole reason good circuitry makes a stone last: not because it draws less, but because less of what it draws goes astray.

The First Law is a consequence of this one. Integrate the rate:

$$\int_0^T \mathbf{w} dt = \int_0^T \eta \cdot (-dQ/dt) dt \leq \int_0^T (-dQ/dt) dt = Q(0) - Q(T) \leq Q(0)$$

Conservation does not need asserting separately. It falls out of efficiency never exceeding one.

Third Law: Impedance

Every material circuit both opposes and modifies what passes through it, so the quality of a Working depends on the material and the geometry together.

$$Z > 0, \text{ always, and } Z = f(M, C, \tau).$$

Impedance is not a property of a substance the way hardness is. Two devices cut from identical steel can differ wildly because one engineer carved a far better circuit, and a superb circuit carved into the wrong material still performs badly. The τ term matters as much: the same circuit that carries a ward cleanly may be poor at carrying fire, so impedance has to be quoted against the working it is meant to carry or it means nothing.

Scholars know impedance can be improved. Nobody has said how far.

What the laws already answer

Three things fall out that the bench does not have the vocabulary to notice.

The bench equation is not a simplification. It is exact, under conditions worth naming. Hold P and Z steady through the cycle and run the stone to exhaustion, and the integral collapses:

$$W = \int_0^T \eta \cdot (-dQ/dt) dt = (P \div Z) \cdot [Q(0) - Q(T)] = Q(0) \cdot P \div Z$$

So the chalked version is the full theory in the case a bench usually cares about, with **Q** meaning **Q(0)**, a fresh stone. It fails in two situations and both are worth knowing. On a partly spent stone the bracket is **Q(0) – Q(T)** and not **Q(0)**. And on a device whose circuit heats or damages as it runs, Z drifts mid-working and cannot be pulled out of the integral — which means the shorthand is least reliable exactly where impedance is already worst. A workshop that trusts the chalk is trusting it hardest on the devices it should trust it least on.

The unknown lower bound is not unknown. Measure P as a fraction of Pmax and Z as a multiple of an ideal circuit, and the Second Law's requirement that $\eta \leq 1$ fixes the floor outright:

$$Z_{\min} = 1$$

A perfect circuit loses nothing. It does not give back more than it was handed. Anything below that floor would mean an apparatus delivering more Working than its stone ever held, which the First Law forbids — so either conservation is wrong or the Third Law's open question was answered the moment the Second Law was written down. Scholars chasing lower impedance are chasing a real gain and an entirely bounded one, and none of them have noticed the bound is already in their own arithmetic.

Which leaves one direction. If efficiency caps at one, no circuit and no gem can raise what a device ultimately delivers past what the stone was handed, and the stone is handed **Qmax** by the charge rune. A better gem closes the gap to Qmax. A better circuit closes a different gap entirely, the one between what the stone gave up and what the device did with it. Neither of them touches Qmax. Every route to a more powerful device short of a better charge rune is a route to the same ceiling approached more tidily — which is, without either academy having framed it that way, an explanation of why twenty years of extraordinary craftsmanship has produced no device more powerful than the first ones out of the ground.

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